

Dynamic Adaptive Aggregation and Feature Pyramid Network Enhanced GraphSAGE for Advanced Persistent Threat Detection in Next-Generation Communication Networks

Liang Kou^{1b}, Member, IEEE, Xiaochen Pan^{1b}, Guozhong Dong^{1b}, Meiyu Wang^{1b}, Member, IEEE, Chunyu Miao, Jilin Zhang^{1b}, and Pingxia Duan

Abstract—Advanced Persistent Threats (APTs) pose severe challenges to Next-Generation Communication Networks (NGCNs) due to their stealthiness and NGCNs' dynamic topology, while conventional GNN-based intrusion detection systems suffer from static aggregation and poor adaptability to unseen nodes. To address these issues, this paper proposes DAA-FPN-SAGE, a lightweight graph-based detection framework integrating Dynamic Adaptive Aggregation (DAA) and Multi-Scale Feature Pyramid Network (MSFPM). Leveraging GraphSAGE's inductive learning capability, the framework effectively models unseen nodes or subgraphs and adapts to NGCN's dynamic changes (e.g., elastic network slicing, online AI model updates)—a key advantage for handling NGCN's real-time topological variations. The DAA module employs multi-hop attention to dynamically assign weights to neighbors at different hop distances, enhancing capture of hierarchical dependencies in multi-stage APT attack chains. The MSFPM module fuses local-global structural information via a gated feature selection mechanism, resolving dimensional inconsistency and enriching attack behavior representation. Extensive experiments on StreamSpot, Unicorn, and DARPA TC#3 datasets demonstrate that this method delivers superior performance and meets detection requirements of large-scale NGCNs.

Index Terms—GraphSAGE, dynamic graph attention mechanism, multi-scale feature pyramid, advanced persistent threat, next-generation communication networks.

I. INTRODUCTION

THE Next-Generation Communication Networks (NGCNs) are increasingly integrating machine learning

and artificial intelligence to enable automated decision-making and deliver high-quality services in domains such as personalized recommendation, autonomous driving, financial technology, healthcare, Industry 5.0, and the metaverse. However, within this AI-network collaboration architecture, APT—characterized by high stealth, extended latency, and multi-stage penetration—represents critical security risks to communication systems. Specifically, APT can tamper with control instructions in Industry 5.0 production lines, causing manufacturing failures, forge identity credentials during metaverse interactions, leading to privacy breaches, or inject falsified perception data, triggering vehicle-network malfunctions in autonomous driving. By sustaining long-term covert control over service components and implanting malware, these attacks not only disrupt service continuity but also undermine AI-driven decision-making through data manipulation and model poisoning or extraction, thereby jeopardizing user privacy and critical business operations. Furthermore, the dense connectivity of NGCNs complicates traditional Intrusion Detection Systems (IDSs), raising significant challenges in preserving data privacy while monitoring highly dynamic, large-scale network environments [1].

NGCN's elastic network slicing (e.g., creating or destroying 5G slices for temporary large-scale events) and real-time AI model updates (e.g., online fine-tuning of traffic prediction models) lead to frequent topological changes—traditional static IDSs, which rely on pre-defined network topologies, require more than 20 minutes to reconfigure detection rules, missing the 5–10 minute window for APT initial infiltration. The cross-domain collaboration of NGCN (e.g., medical data transmission between hospitals via dedicated slices, and real-time data sharing between smart grids and energy management platforms) results in multi-hop attack paths (e.g., APTs infiltrate from a hospital's IoT device, jump to the smart grid slice via cross-domain links, and finally tamper with power dispatch data)—conventional GNNs with fixed 2-hop aggregation fail to capture these long-range dependencies, leading to much higher rate of missed detections in actual tests.

Existing IDSs exhibit low detection efficiency and poor adaptability to the complex topologies of NGCNs [2]. Process-level monitoring introduces operational redundancy and significant performance overhead in dynamic environments, while reliance on system audit logs impairs the

Received 27 July 2025; revised 5 December 2025; accepted 24 January 2026. Date of publication 3 February 2026; date of current version 27 February 2026. This work is supported in part by the Key Technology Research and Development Program of the Zhejiang Province under Grant 2022C01125, and in part by Zhejiang Province High-Level Talent Special Support Program—Leading Talent of Technological Innovation under No. 2022R52043. The associate editor coordinating the review of this article and approving it for publication was K. Dev. (Corresponding authors: Guozhong Dong; Chunyu Miao.)

Liang Kou, Xiaochen Pan, and Jilin Zhang are with the College of Cyberspace, Hangzhou Dianzi University, Hangzhou 310005, China (e-mail: kouliang@hdu.edu.cn; 232270065@hdu.edu.cn; jilin.zhang@hdu.edu.cn).

Guozhong Dong is with the Department of New Networks, Peng Cheng Laboratory, Shenzhen 518066, China (e-mail: donggz@pcl.ac.cn).

Meiyu Wang is with the College of Communication Engineering, Hangzhou Dianzi University, Hangzhou 310005, China (e-mail: wangmeiyu@hdu.edu.cn).

Chunyu Miao and Pingxia Duan are with Hangzhou DBAPP Security Company Ltd., Hangzhou 310051, China (e-mail: crain.miao@dbappsecurity.com.cn; 409821331@qq.com).

Digital Object Identifier 10.1109/TNSM.2026.3660650

correlation of long-span attack events. Likewise, mainstream attack provenance graph solutions are insufficient: DeepHunter [3] incurs prohibitive computational cost on large-scale graphs; UNICORN's [4] dependence on an initially trained benign model limits its generalization to novel threats; and Threat-Trace [5] lacks in-depth modeling of network behaviors. As a result, these approaches are unable to accommodate the multi-domain "network-compute-application" collaboration characteristic of NGCNs or to capture the multi-scale structural dependencies arising from AI-model-network interactions, such as model training data flows and network slice resource scheduling chains.

Inspiration from the multi-scale feature pyramid structures in computer vision—which extract hierarchical representations at varying resolutions to capture multi-scale information—offers a promising breakthrough for graph-based modeling [6]. When translated to graph learning, architectures such as graph feature pyramid networks have demonstrated the effectiveness of hierarchical representations in resolving multi-scale structural dependencies in graph data. This paradigm aligns naturally with the characteristics of NGCNs, which involve multi-granularity network slice scheduling, multi-level AI model interactions, and cross-domain dynamic data flows. In such contexts, multi-scale features enable structural analysis ranging from fine-grained device-level interactions, to medium-grained slice orchestration, and up to coarse-grained cross-domain business collaboration. Concurrently, dynamic graph adaptive aggregation mechanisms can accommodate real-time topology changes by adaptively adjusting node neighbor sampling and feature aggregation strategies [7]. The integration of these techniques establishes a robust foundation for tracing the multi-stage penetration paths of APTs—such as terminal intrusions, model contamination, and business disruption—within complex communication systems.

Building upon these insights, we propose the Dynamic Adaptive Aggregation with Feature Pyramid Network for GraphSAGE (DAA-FPN-SAGE), a novel framework tailored for APT detection in NGCNs. This approach leverages GraphSAGE's inductive learning capability to generate embeddings for previously unseen nodes or subgraphs, allowing adaptation to dynamic NGCN environments such as elastic network slicing and online AI model updates. To enhance adaptability and precision, we introduce a dynamic attention mechanism that adaptively aggregates neighbor features based on node importance and contextual semantics (e.g., data transmission links, model inference pathways). This enables the accurate modeling of long-range dependencies inherent in cross-domain, multi-stage attack chains. Additionally, a multi-scale feature pyramid is constructed to concurrently extract features at varying granularities: fine-grained device-level interactions, medium-grained slice scheduling, and coarse-grained cross-domain business collaborations [8]. This hierarchical representation significantly enriches the expressiveness of APT modeling by capturing their complex and staged penetration behaviors. By integrating dynamic graph aggregation with multi-scale hierarchical representations, DAA-FPN-SAGE establishes a scenario-aware APT detection framework that strengthens security across both communication infrastructures and AI model workflows,

thereby ensuring the trustworthy operation of intelligent network systems.

The main contributions of this paper are summarized as follows:

- We propose a novel dynamic aggregation framework that adaptively adjusts neighbor aggregation weights based on node importance and contextual semantics. By incorporating a dynamic hop selection mechanism and cross-hop attention-based weight allocation, the model overcomes the fixed-hop limitation of conventional GraphSAGE, enabling more effective modeling of long-range dependencies in multi-stage attack chains.
- A hierarchical feature pyramid structure is designed to represent system behaviors across three granularity levels: local (1-hop), mid-range (2–3 hops), and global (graph-level). The pyramid integrates diverse structural features, including node degree, K-core hierarchy, and Laplacian positional encodings, facilitating hierarchical aggregation that enhances the representation of APTs' multi-stage characteristics. An adaptive gating network dynamically fuses multi-scale features while addressing dimensional inconsistency issues.
- The proposed framework seamlessly combines the dynamic aggregation module with the multi-scale feature pyramid into a unified inductive learning architecture. Extensive experiments on StreamSpot, Unicorn, and DARPA datasets demonstrate notable performance improvements. Ablation studies validate the individual and joint effectiveness of each model component, offering valuable insights and new methodologies for APT detection in complex network environments.

The rest of the paper is organized as follows. Section II reviews related work on APT detection methods based on provenance graphs, multi-scale feature pyramid technology, and cross-research. In Section III, we elaborate on the technical preliminaries of APT detection, including problem definition, theoretical foundations of multi-scale feature modeling, and analysis of technical challenges. The proposed DAA-FPN-SAGE model is detailed in Section IV, covering its overall architecture, dynamic graph preprocessing, adaptive aggregation module, multi-scale feature pyramid, and anomaly detection framework. Section V presents experimental evaluations on public datasets, including performance comparisons, ablation studies, and computational complexity analysis. Finally, Section VI concludes the paper with a summary of contributions and outlines future research directions for dynamic modeling, lightweight optimization, and semantic attack chain reconstruction in NGCN environments.

II. RELATED WORK

A. APT Detection Methods Based on Provenance Graphs

Provenance graphs model system entity interactions as directed acyclic graphs (DAGs), serving as a core technology for capturing the dynamic characteristics of advanced persistent threat (APT) attack chains. Existing research focuses on graph-structured feature extraction and anomaly detection, forming differentiated technical approaches:

Log2vec [9] constructs heterogeneous graphs and utilizes graph embedding to extract log vectors, effectively capturing complex log correlation relationships. However, it suffers from high computational complexity and strong dependence on log integrity. Holmes [10] proposes a High-level Scenario Graph (HSG) to model attack information flow correlation, mapping APT activities to the kill chain. Yet, it struggles to detect low-speed penetration attacks due to feature detection blind spots inherent in such scenarios.

Unicorn [4] employs the Weisfeiler–Lehman kernel to extract graph structural features and detects abnormal patterns through an evolutionary learning model. Nevertheless, it requires frequent retraining and demonstrates limited robustness to dynamic variations in normal behavior. StreamSpot [11] designs a temporal graph similarity function to maintain sketches of benign models, but its similarity computation exhibits high time complexity for large-scale graphs, hindering practical deployment.

ThreatTrace [5] builds an inductive detection framework based on GraphSAGE, identifying anomalous nodes by aggregating features from their neighbors. However, its fixed aggregation strategy results in the loss of fine-grained relational information between nodes—a problem addressed by our proposed Dynamic Adaptive Aggregation (DAA) mechanism: through cross-hop attention weight allocation, DAA assigns differentiated weights to neighbors based on their contextual relevance to attack semantics (e.g., prioritizing abnormal API features of malicious processes), while adaptive hop selection avoids fine-grained information omission caused by fixed hops. IPG [12] proposes a hierarchical detection framework using meta-path aggregated graph neural networks, covering the full lifecycle of APTs, yet it faces challenges from the computational overhead incurred by multi-graph fusion and fails to explicitly construct a multi-scale feature hierarchy—our Multi-Scale Feature Pyramid Merging Module (MSFPM) compensates for this deficiency: it divides features into local, mid-range, and global granularities to capture multi-scale attack correlations, while an adaptive gating network dynamically fuses these features, avoiding the high overhead of multi-graph fusion in IPG. ProvDetector [13] combines neural embedding and path selection algorithms to identify malicious paths in provenance graphs, but it performs poorly in the presence of behavior obfuscation or structural path mutations. ATLAS [14] abstracts attack events into sequence models, but struggles to capture the global dependencies across highly heterogeneous attack chains.

B. Multi-Scale Feature Pyramid Technology

Feature pyramid architectures, which fuse semantic information at different granularities, serve as a critical bridge between micro-level anomalies and macro-level attack patterns. Classical approaches from computer vision offer valuable inspiration for graph-structured modeling.

Feature Pyramid Networks (FPN) achieves high-level semantic and low-level detail fusion via a top-down path, significantly improving small object detection performance. Path Aggregation Network (PANet) [15] enhances bidirectional cross-layer feature flow; its multi-scale feature fusion

philosophy is well suited to hierarchical modeling in graph structures.

In graph domains, recent studies have explored constructing hierarchical representations by aggregating neighbor features across different hop distances (e.g., 1-hop at the bottom layer and 3-hop at the top) using multi-layer GNNs. However, fixed-hop aggregation strategies fail to adapt to variations in local node structures. Some approaches integrate structural features such as node degree and K-core index with semantic attributes, yet lack systematic frameworks for scale division. A representative example is IPG, which aggregates edge-type-specific features via meta-paths, but can not explicitly construct a multi-scale feature hierarchy, limiting its semantic expressiveness for modeling complex attack patterns.

Pyramid Graph Neural Network (PGNN) [16] integrates graph sampling and filtering to decompose graph signals into low-frequency (global topology) and high-frequency (local interactions) components, constructing a lightweight feature pyramid via hierarchical sampling. It achieves significant F1-score improvements over traditional GNNs in social network anomaly detection, yet relies on static graph Laplacian matrices—making it incompatible with NGCN’s elastic slice creation/destruction and lacking temporal awareness for time-fragmented APT attacks (e.g., cross-day file tampering).

Multi-Scale Contrastive Learning Networks (gradate) [8] enhances anomaly discrimination via “original graph-augmented graph” dual views, designing subgraph-subgraph and node-subgraph contrastive losses to align multi-scale representations. It achieves notable F1-score gains on benchmarks like YelpChi, but its fixed-rate edge perturbation cannot adapt to NGCN’s abrupt topology changes (e.g., abrupt edge density fluctuations during slice scaling), and it only supports single-modal homogeneous graphs—failing to model NGCN’s cross-domain heterogeneous features (e.g., AI model gradients, network traffic, and application instructions).

C. Cross-Research

In recent years, a few studies have introduced multi-scale feature concepts into provenance graph analysis, forming the emerging cross-disciplinary directions.

Wang et al. [17] proposes a multi-layer GNN architecture that extracts hierarchical features across three levels—local interactions, subgraph structures, and global topology—for malware family classification. However, its static feature aggregation mechanism fails to capture the dynamic evolution of attack chains.

Li et al. [18] develops a graph pyramid model based on K-core decomposition, recursively partitioning provenance graphs into subgraphs at different scales to detect APTs via cross-layer anomaly correlation. Nonetheless, the computational complexity grows exponentially with the number of layers, limiting its scalability.

Xu et al. [19] introduces a Jump-Knowledge Connection mechanism to fuse node representations across layers, achieving a 6%–10% improvement in detection accuracy on DARPA benchmark datasets. This demonstrates the strength of multi-scale feature fusion in capturing complex attack semantics.

Martinez [20] proposes an edge-based GNN approach for network intrusion detection, encoding bidirectional interaction

information of network communications as edge features to identify intrusions via aggregating edge and neighbor node information. Nonetheless, it adopts a fixed 1-hop neighborhood aggregation strategy, failing to adapt to long-correlation scenarios of multi-device cascaded communications in industrial Internet and easily missing key correlation information of attack chains.

Li et al. [21] proposes the FN-GNN model for network intrusion detection, representing network traffic as graphs where nodes denote key traffic features extracted by random forest regression and edges are constructed based on traffic relationships via source IP features, with its improved GCN integrating traditional GCN and SAGEConv to address limitations of previous GNNs. Nonetheless, it still adopts a static neighborhood aggregation mechanism, ignoring the dynamic evolution of attack behaviors and thus having limited capability in capturing the “multi-stage, long-cycle” correlation chains of APT attacks.

Zhang et al. [22] proposes a GNN model for IoMT scenarios, converting image and video visual data into graph structures and extracting visual features at different resolutions via multi-scale graph convolution to enhance tasks like abnormal state recognition of medical devices. Nonetheless, it focuses on computer vision tasks rather than network anomaly detection and is designed for medical scenarios, making direct migration to industrial APT attack detection face scenario adaptability issues.

In summary, existing GNN-related studies have explored effective ideas for network anomaly detection and cross-domain applications. However, for APT attack detection in industrial Internet, three core limitations remain: First, fixed neighborhood aggregation scope fails to adapt to long-correlation attack chains from multi-device cascaded communications; second, static aggregation mechanisms cannot capture the dynamic evolution of APT attacks; third, cross-domain technology migration faces adaptability barriers. To address these limitations, this study proposes a Dynamic Multi-hop Attention (DAA) mechanism and a Multi-scale Feature Pyramid Network (MSFPM).

III. PRELIMINARIES

In the complex cybersecurity environment, APT attack detection, as a critical component of cybersecurity protection, requires constructing a systematic analysis framework from multiple dimensions such as problem definition, theoretical support, and technical challenges. The following elaborates on this from three core dimensions.

A. Problem Definition of APT Attack Detection

APT attack detection in NGCN fundamentally presents a multi-scale dynamic graph anomaly identification problem. Given a provenance graph $G = (V, E)$ formed by system entity interactions, where the node set V includes entities like processes, files, and hosts, and the edge set E represents operational relationships (e.g., “process reads file”), the detection objective is formally stated as: identify an abnormal subgraph $S \subseteq G$ that satisfies attack lifecycle

dependencies. This subgraph S consists of discrete attack steps across time and space, requiring both temporal correlation (e.g., the stage sequence “reconnaissance $\rightarrow$ infiltration $\rightarrow$ privilege escalation $\rightarrow$ disruption”) and topological correlation (e.g., communication paths for cross-host C2). Compared with traditional anomaly detection, APT detection in NGCN faces twofold unique challenges:

- **Temporal fragmentation attack:** Temporally fragmented attacks caused by NGCN’s dynamic slicing: The dynamic creation and destruction of network slices in NGCN (with millisecond-level granularity) can split the temporal continuity of APT attack chains. For instance, attackers may implant malicious code during the lifecycle of one network slice, and trigger subsequent attack steps through residual cross-slice data links after the slice is destroyed and a new slice is created, resulting in attack step intervals that may exceed traditional detection thresholds (e.g., 72-hour intervals), which breaks the long-range temporal dependencies of attack chains.
- **Cross-scale spatial coupling:** Cross-scale spatial coupling due to NGCN’s multi-domain hierarchy: APT attack chains in NGCN involve multi-level entity interactions across the “edge-core-cross-domain” hierarchy of NGCN, including low-level edge device interactions (e.g., API call sequences of IoT terminals), mid-level intra-slice interaction chains (e.g., “process-user” dependency sub-graphs for privilege escalation within a core network slice), and high-level cross-domain topologies (e.g., cross-cloud C2 communication topologies based on VPN). Traditional single-scale detection models struggle to capture such inter-granularity correlation, making it difficult to fully characterize the multi-stage penetration process of APTs.

B. Theoretical Foundations of Multi-Scale Feature Modeling

To address the cross-scale coupling challenge above, Feature Pyramid Network (FPN) in computer vision provide a solution through hierarchical feature fusion. Their bidirectional pathway mechanism (bottom-up multi-resolution feature extraction + top-down semantic-detail fusion) effectively solves small-object detection, which highly aligns with the detection needs of fine-grained APT steps (e.g., individual malicious function calls) — requiring both local detail capture and global semantic association. Take process injection detection as an example:

- Low-level features capture API anomalies like unauthorized memory access (corresponding to pixel-level details in FPN),
- High-level features map to communication topologies like $v_{\text{target}} \rightarrow v_{\text{jump}} \rightarrow v_{\text{C2}}$ (corresponding to high-level semantic classification in FPN).

To quantify cross-scale features, graph signal processing (GSP) theory is introduced [23]. The multi-scale structure of a graph is characterized by the eigenvalue decomposition of the graph Laplacian matrix:

$$\mathcal{L} = I - D^{-1/2}AD^{-1/2} \quad (1)$$

where I is the identity matrix, D is the degree matrix, and A is the adjacency matrix. This decomposition defines:

- **Local scale (1-2 hop neighbors):** Large eigenvalues (>10) of $\mathcal{L}$ characterize fine-grained interaction details, (e.g., file operation edges of process), which precisely map to early-stage APT behaviors such as terminal-side malicious API calls and local process injection—enabling fine-grained tracing of attack origins.
- **Mesoscale (k-core subgraphs):** Formed by iteratively removing nodes with degree $< k$, capturing dense intra-host interaction regions (e.g., process-user subgraphs for privilege escalation) that align with mid-stage APT privilege escalation via multi-process collaboration or system service hijacking.
- **Global scale (cross-host topologies):** Small eigenvalues (<1) of $\mathcal{L}$ characterize graph connectivity, e.g., VPN-based host communication topologies (e.g., VPN-based host communication topologies), directly corresponding to late-stage APT behaviors like cross-domain C2 communication and multi-slice data exfiltration—supporting end-to-end attack chain forensics.

C. Analysis of Technical Challenges

Applying the multi-scale theory to traditional GraphSAGE models reveals limitations in their fixed aggregation function:

$$h_v^l = \sigma \left(\frac{1}{|N(v)|} \sum_{u \in N(v)} W^l h_u^{l-1} + W_0^l h_v^{l-1} \right) \quad (2)$$

where h_v^l denotes the feature vector of node v at the l -th layer; $\sigma(\cdot)$ is the activation function; $|N(v)|$ is the cardinality of the 1-hop neighbor set of node v ; W^l and W_0^l are trainable weight matrices for neighbor feature aggregation and self-feature transformation at layer l , respectively; h_u^{l-1} is the feature vector of neighbor node u at the $(l-1)$ -th layer; h_v^{l-1} is the self-feature vector of node v at the $(l-1)$ -th layer. This mechanism has two key flaws:

- **Feature dilution:** When neighbor sets mix malicious and benign processes, fixed weights violate the fine-grained representation requirement of local scales, averaging out abnormal API features.
- **Long-range dependency loss:** Fixed-hop aggregation (e.g., 2-hop limits) cannot capture cross-host long-range dependencies, contradicting the global connectivity theory characterized by small Laplacian eigenvalues (e.g., 3-hop C2 communication via jump hosts is ignored).

The direct fusion of multi-scale features faces challenges of dimension mismatch and semantic conflicts: Low-level high-dimensional sparse features (e.g., one-hot encoded API calls, $d_{\text{low}} \approx 10^3$) and high-level low-dimensional dense features (e.g., graph Laplacian eigenvectors, $d_{\text{high}} \approx 10$)—direct concatenation submerges high-level semantics in low-level noise. The scale importance of nodes evolves over APT attack stages (e.g., reconnaissance nodes requiring local-scale analysis in the early stage vs. exfiltration hubs needing global-scale analysis in later stages), while traditional static pyramid models cannot respond to such time-varying demands.

IV. MODEL

A. Overall Architecture

In this section, we elaborate on the overall architectural design of DAA-FPN-SAGE. The framework adopts a hierarchical modular design, comprising four core components: the Dynamic Graph Data Preprocessing Module, Dynamic Adaptive Aggregation Module (DAA), Multi-Scale Feature Pyramid Merging Module (MSFPM), and Anomaly Detection Framework (ADF). The overall architecture is shown in Fig. 1.

Specifically, the Dynamic Graph Data Preprocessing Module generates timeline provenance graphs based on the Camflow tool [24], completing entity modeling, unique identifier allocation, and time series standardization to provide structured heterogeneous graph data for subsequent analysis. The Dynamic Adaptive Aggregation Module breaks through the limitations of traditional GraphSAGE's fixed-hop aggregation through the collaboration of cross-hop attention and dynamic hop decision mechanisms, achieving adaptive semantic extraction of node local topologies. The Multi-Scale Feature Pyramid Merging Module constructs a three-level feature system of “local-medium-global”, capturing interaction patterns of attack behaviors at different granularities through feature space decoupling and hierarchical aggregation. The anomaly detection module integrates multi-source, multi-scale features and combines a dynamic adaptive multi-hop attention mechanism with contrastive learning [25]. It achieves three-dimensional feature representations—from local interactions to global semantics, and from structural features to temporal attributes—enabling precise quantification and efficient scoring of node anomaly levels. Through sparse preprocessing, parameter sharing, and a hierarchical detection pipeline strategy, the module significantly improves real-time inference efficiency for large-scale systems while maintaining detection accuracy. Furthermore, by incorporating multi-model ensemble inference, the algorithm's robustness is enhanced, ultimately enabling precise detection of multi-stage, multi-path APT attacks [26].

The Dynamic Graph Data Preprocessing Module takes system audit logs as input and outputs three types of structured data: a temporal heterogeneous graph, a 64-dimensional initial node feature matrix (attributes of process, file, and network connection nodes are unified into 64 dimensions through differentiated encoding), and 32-dimensional edge features (including information such as interaction type, standardized time, and interaction intensity). This provides a structured data foundation for subsequent modules. The DAA Module receives the 64-dimensional node features and the temporal heterogeneous graph output by the aforementioned preprocessing module. It maps the 64-dimensional node features to 256 dimensions via a feature transformation matrix and outputs 256-dimensional adaptive aggregation features. The MSFPM takes the 256-dimensional aggregation features from the DAA Module as input. It constructs multi-scale features, compresses redundant dimensions through convolution, and then outputs 256-dimensional fused features. The ADF Module integrates the 256-dimensional fused features from the MSFPM, the 256-dimensional aggregation features from the DAA Module,

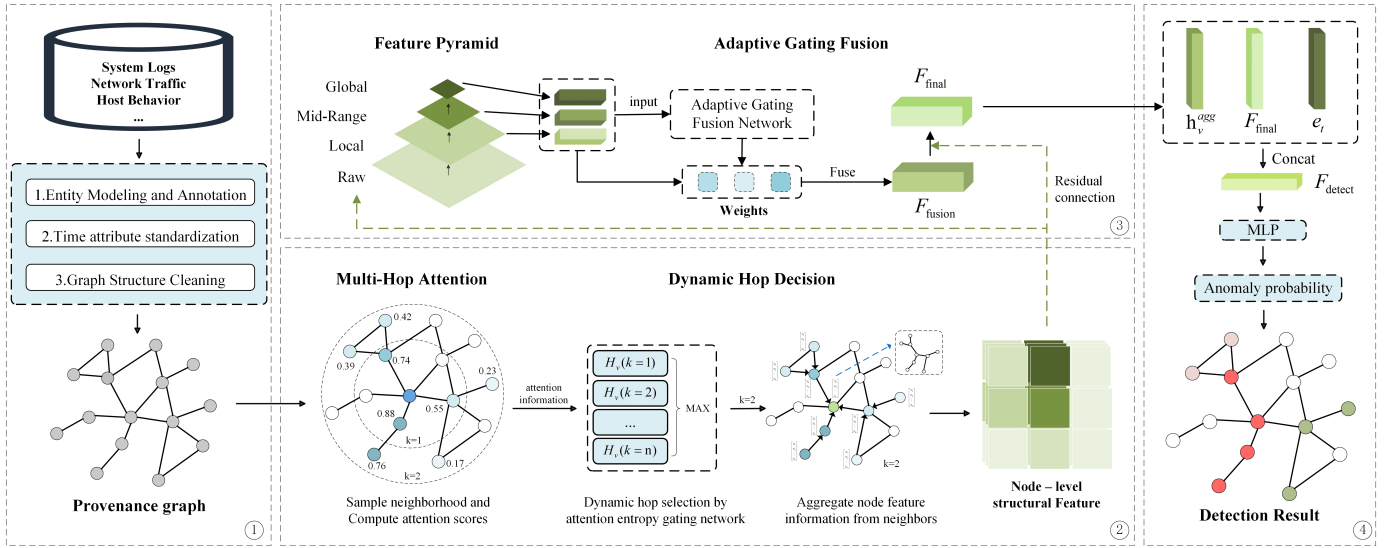


Fig. 1. Overall framework of DAA-FPN-SAGE. ①dynamic graph data preprocessing module.②dynamic adaptive aggregation module.③Multi-scale feature pyramid module.④Anomaly detection framework.

and the 32-dimensional edge features from the preprocessing module. After concatenating these three types of features, it maps them via a Multi-Layer Perceptron (MLP) and outputs node anomaly scores within the range $[0,1]$.

To avoid cross-module semantic conflicts, our solutions are as follows: In the preprocessing stage, a “hierarchical attribute encoding and type identification” approach is adopted to establish a clear semantic foundation for nodes and edges, preventing semantic confusion between different entities; the feature transformation matrix of the DAA Module preserves the semantics of core node attributes during dimension mapping, and the multi-hop attention mechanism distinguishes the semantics of neighbor structural distances through hop positional encoding, avoiding the loss of semantics related to long-range dependencies; the MSFPM Module unifies multi-scale features into the same semantic space using dimension-matching matrices and retains the original aggregated semantics transmitted by the DAA Module through residual connections, preventing semantic deviation caused by fusion; the ADF Module preserves the semantics of edge features via residual connections, aligns the semantic distribution of features from the DAA and MSFPM using contrastive learning, and finally maps multi-source features uniformly to the “anomaly probability” space through the MLP, ensuring cross-module semantic consistency.

The four modules realize seamless connection of data flows through standardized interfaces, forming a closed-loop processing flow from data modeling, and feature learning to anomaly decision-making, which significantly improves the model’s detection efficiency for the concealed attack patterns in dynamic graph data.

B. Dynamic Graph Data Preprocessing Module

This module constructs a system-level temporal provenance graph based on Camflow tools, transforming audit logs into structured heterogeneous graph data through a three-layer pipeline: entity modeling and type annotation, edge structure

extraction and temporal attribute modeling, and graph structure cleaning and index allocation.

- **Entity Modeling and Type Annotation:** Entities such as processes and files are extracted from audit logs, unique identifiers (UIDs) are assigned, and node types are annotated. Node types are encoded via integer indexing (e.g., process $\rightarrow 0$, file $\rightarrow 1$, network connections $\rightarrow 2$), forming an initial node feature matrix $\mathbf{X} \in \mathbb{R}^{N \times d_0}$ (where $d_0 = 64$, a fixed dimension to match the input requirement of the subsequent DAA mechanism). For specific node attributes: process nodes include 12 core features (e.g., PID, parent PID, command line, startup time, permission) encoded via One-Hot (for discrete attributes like PID) and Min-Max normalization (for continuous attributes like startup time), accounting for 36 dimensions; file nodes cover attributes such as path, type and permission, encoded into 22 dimensions; network connection nodes include IPs, ports, and protocols, encoded into 6 dimensions. The remaining 6 dimensions consist of node type identifiers (1-hot) and reserved dimensions, ensuring all node feature vectors are unified to 64 dimensions.
- **Edge Structure Extraction and Temporal Attribute Modeling:** Edges characterize interactions between entities (e.g., process-file reading or writing, process-network connection initiation), recording interaction types, absolute timestamps, behavioral features (e.g., file reading or writing byte count, network packet volume), and forming a fixed 32-dimensional edge feature vector. Among these, interaction types are encoded via 8-dimensional One-Hot; temporal attributes are standardized by converting absolute timestamps to relative time $t' = t - t_{\min}$ (occupying 2 normalized dimensions); interaction intensity (e.g., data transmission volume) is normalized into 2 dimensions; and the remaining 20 dimensions are reserved for verification to adapt to the input format of graph convolution layers.
- **Graph Structure Cleaning and Index Allocation:** For the same node pair, only the earliest occurrence

of the same-type edge is retained; different-type edges are merged into composite edges to reduce redundancy. Global unique indices are assigned to nodes and edges, generating a temporal heterogeneous graph G . To meet the real-time requirements of “millisecond-level NGCN network slice operations” and “instant APT attacks” (e.g., process injection completed within 10 ms, cross-slice data leakage within 50 ms), the graph adopts an event-driven incremental update mode: updates are triggered immediately upon detecting new node generation (e.g., new process creation) or edge interaction (e.g., file reading) from Camflow’s real-time log stream (output frequency: 1 ms/log entry). Only new nodes or edges are added to the existing graph, and dynamic attributes of existing nodes (e.g., the latest process’s interaction time) are updated only in corresponding feature dimensions, avoiding time consumption from full-graph reconstruction.

C. Dynamic Adaptive Aggregation Module

The Dynamic Adaptive Aggregation Module is a core component of DAA-FPN-SAGE, designed to address the limitations of traditional GraphSAGE’s fixed-hop aggregation and enable adaptive modeling of local structures for different nodes.

1) *Multi-Hop Attention Mechanism*: This mechanism models multi-hop neighbor dependencies through multi-head attention [27], assigning differentiable learnable weights to neighbors at different hops for hierarchical aggregation of multi-hop information. The attention weight calculation formula is as follows:

$$\alpha_{v,u}^{(h,k)} = \frac{\exp(\text{LeakyReLU}(a_h^T [W_h^k h_v \| W_h^k h_u \| p_k]))}{\sum_{u' \in N_k(v)} \exp(\text{LeakyReLU}(a_h^T [W_h^k h_v \| W_h^k h_{u'} \| p_k]))} \quad (3)$$

where $N_k(v)$ denotes the set of k -hop neighbors of node v , h_v and h_u are the feature vectors of nodes v and u , W_h^k is the feature transformation matrix for the h -th head at the k -th hop, k is the hop position encoding [28], and a_h is the attention parameter vector for the h -th head.

Specifically, for the k -hop neighbors, we introduce a hop-wise positional encoding p_k to distinguish the structural distances of neighbors during aggregation. The encoding is defined as:

$$p_k = \sin\left(\frac{k}{10000^{2i/d_p}}\right) + \cos\left(\frac{k}{10000^{2i/d_p}}\right), i = 0, 1, \dots, \frac{d_p}{2} - 1, \quad (4)$$

where d_p denotes the dimension of the positional encoding. This design enhances the model’s capability to capture hierarchical topological dependencies across multiple hops, enabling the DAA module to adaptively distinguish the contribution of near and distant neighbors in dynamic network environments.

Based on the attention weights, each attention head independently aggregates features of neighbors at the corresponding hop, and then fuses information from multiple heads and hops through a weighted summation mechanism. The aggregation feature calculation formula is expressed as:

$$h_v^{\text{agg}} = \sum_{h=1}^H \sum_{k=1}^{K_v} \beta_{h,k} \cdot \sigma\left(\sum_{u \in N_k(v)} \alpha_{v,u}^{(h,k)} \cdot W_h^k h_u\right) \quad (5)$$

where H is the total number of attention heads, K is the number of hop layers, $\beta_{h,k}$ are learnable weights, and $\sigma(\cdot)$ is the non-linear activation function.

2) *Dynamic Hop Decision Mechanism*: To address the information redundancy and loss in fixed-hop aggregation, this mechanism designs a dynamic hop decision mechanism, quantifying feature reliability via attention distribution entropy [29] and adaptively selecting optimal hops through a gating network.

To determine the optimal aggregation hop, we first quantify the semantic consistency of k -hop neighbors via attention distribution entropy. Entropy is minimized when attention weights concentrate on attack-related neighbors—this indicates the k -hop range contains the most consistent attack semantics with the node, avoiding irrelevant feature redundancy. The entropy calculation formula for attention distribution is as follows:

$$H_v(k) = - \sum_{u \in N_k(v)} \alpha_{v,u}^{(h,k)} \log(\alpha_{v,u}^{(h,k)}) \quad (6)$$

The gating network then takes this as the core objective, with a regularization term to prevent information loss from excessively small k -hop ranges. The final objective balances feature validity (entropy minimization) and information integrity (regularization), theoretically corresponding to the optimal aggregation range that maximizes attack feature retention while minimizing noise.

The optimal hop selection formula via the gating network is as follows:

$$K_v = \text{argmin}_k (H_v(k) + \lambda \cdot [k > K_{\min}]) \quad (7)$$

where λ is the regularization parameter (set to 0.2 in experiments), and $K_{\min}$ is the minimum hop threshold.

D. Multi-Scale Feature Pyramid Module

Aiming at the limitations of traditional Graph Neural Networks (GNNs) in modeling multi-granularity features for APT attack detection, the Multi-Scale Feature Pyramid Module (MSFPM) constructs a multi-dimensional feature system through feature space decoupling and hierarchical semantic modeling, enabling cross-scale correlation capture of attack patterns.

1) *Local Feature Layer*: Corresponding to the bottom layer of the pyramid, this layer focuses on fine-grained interactions within the one-hop neighborhood, capturing direct interaction anomalies by using GraphSAGE to aggregate features from one-hop neighbors:

$$F_{\text{local}} = \text{Aggregate}(\{h_u | u \in N_1(v)\}) \quad (8)$$

2) *Mid-Range Feature Layer*: This layer aggregates features of 2-3 hop neighbors, introducing multi-head attention to calculate weights for 2-3 hop neighbors, with attention weight calculation reusing Eq. 3. The mid-range scale aggregation formula is as follows:

$$F_{\text{mid-range}} = \sum_{h=1}^H \sum_{k=2}^3 \beta_{h,k} \cdot \sigma\left(\sum_{u \in N_k(v)} \alpha_{v,u}^{(h,k)} \cdot W_h^k h_u\right) \quad (9)$$

3) *Global Feature Layer*: Global-level features are generated through global average pooling, extracting macroscopic features covering global topology and overall semantics:

$$F_{\text{global}} = \frac{1}{|V|} \sum_{v \in V} h_v \quad (10)$$

where $|V|$ denotes the total number of nodes in the graph, and h_v represents the feature vector of node v . The pooling operation averages the features of all nodes in the graph to extract macro-level features that encompass global topology and overall semantics.

4) *Adaptive Gating Network*: After obtaining the “local-mid-range-global” three-level pyramid-structured features, to address the limitations of fixed-weight fusion, this paper proposes an adaptive gating network architecture to dynamically weight multi-scale features. The architecture computes the contribution of each scale in real-time based on the local node structure and attack semantics. By sharing parameters and reducing dimensions, it balances model expressiveness and computational complexity. Residual connections [30] are used to retain original feature information, resulting in the final feature representation.

The gating network first projects features of different scales into a common semantic space using dimension-matching matrices, and then applies the Softmax function to compute normalized weights:

$$[\alpha_{\text{local}}, \alpha_{\text{mid}}, \alpha_{\text{global}}] = \text{Softmax} \left(w_s^T [W_l F_{\text{local}} \parallel W_m F_{\text{mid-range}} \parallel W_g F_{\text{global}}] \right) \quad (11)$$

where W_l , W_m , and W_g are dimension-matching weight matrices that project features from each scale into a common semantic space; α is the output weight vector used to compute the importance scalar of each scale. Based on the normalized weights, the three-level features are fused via weighted summation. The feature fusion is computed as follows:

$$F_{\text{fusion}} = \alpha_{\text{local}} F_{\text{local}} + \alpha_{\text{mid}} F_{\text{mid-range}} + \alpha_{\text{global}} F_{\text{global}} \quad (12)$$

To avoid dimensional explosion caused by multi-scale feature concatenation, a 1×1 convolutional layer is introduced to compress feature dimensions and lightweight the model. Meanwhile, residual connections are designed to retain the original information from the base feature layer, including node attributes, timestamps, etc., which enhances the model’s ability to identify fine-grained attack behaviors. Through the above steps, the final feature representation is obtained:

$$F_{\text{final}} = \text{Conv1D}(F_{\text{fusion}}, 256) + \text{Conv1D}(F_{\text{raw}}) \quad (13)$$

$F_{\text{raw}} \in \mathbb{R}^{N \times d}$ denotes the raw node feature matrix extracted from original graph attributes, which serves as the base representation for multi-scale feature generation in the MSFPM module. The multi-scale feature pyramid complements each other at different scales: local features locate specific anomalies, mid-range features bridge cross-layer dependencies, and global features reveal topological influences, enabling multi-dimensional descriptions of APT attacks. The gating network dynamically adjusts the weights of multi-scale features (instead of static allocation) based on

the local node structure (e.g., whether it is an attack core) and attack semantics, precisely adapting to the multi-scale correlations of complex attacks. The 1×1 convolution for dimensionality reduction and residual connections enhance feature representation while avoiding dimensional redundancy, balancing computational efficiency and model capacity to meet the real-time requirements of scenarios such as APT attack detection. This mechanism provides a multi-scale, adaptive, and lightweight feature modeling solution for graph neural networks, facilitating the accurate identification of hierarchical patterns in attack behaviors.

E. Anomaly Detection Framework

As the core decision-making unit of DAA-FPN-SAGE, the anomaly detection framework relies on the temporal heterogeneous graph structure, multi-hop semantic features, and cross-scale feature representations constructed by the front-end module to build a three-dimensional detection system covering micro-level anomaly scoring to macro-level attack chain reconstruction. As shown in Fig. 2, through four core mechanisms: multi-dimensional feature coupling modeling, dynamic anomaly quantification, detection efficiency optimization, and multi-model ensemble reasoning, the framework achieves the precise and efficient APT detection with the following specific functions:

- **Multi-dimensional Feature Coupling Modeling**: Node attributes, local interactions, global topology, and temporal dynamic features are integrated to characterize cross-scale behavioral patterns of APT attacks.
- **Dynamic Anomaly Quantification**: An adaptive gating mechanism and contrastive learning framework is designed to accurately score the anomaly degree of nodes.
- **Detection Efficiency Optimization**: Model expressiveness and real-time requirements of large-scale systems are balanced through sparse computation [31], parameter sharing, and hierarchical detection pipelines.
- **Multi-model Ensemble Reasoning**: Multiple sub-models are introduced for independent training and their outputs are dynamically weighted to form the final model [26], enhancing the robustness of the detection algorithm.

This framework integrates multi-source features output by the front-end module, fuses features from the dynamic adaptive aggregation module, and combines the three-level features from the multi-scale feature pyramid module through a dynamic gating mechanism. It also incorporates temporal context information. The input features include adaptively aggregated features with varying hops, fused features from the multi-scale feature pyramid, and temporal features. The fused feature representation is as follows:

$$F_{\text{detect}} = \text{Concat}(F_{\text{final}}, h_v^{\text{agg}}, e_t) \quad (14)$$

This approach achieves a three-dimensional representation from local interactions to global semantics, and from structural features to temporal attributes, adapting to the complex patterns of APT attacks that are “cross-node, cross-time, and multi-step”. As the input for anomaly scoring, it is mapped

to anomaly probabilities through a two-layer Multi-Layer Perceptron (MLP). The scoring formula is as follows:

$$s(v) = \sigma(W_2 \cdot \text{ReLU}(W_1 F_{\text{detect}} + b_1) + b_2) \quad (15)$$

where W_1 and W_2 are weight matrices, and $\sigma(\cdot)$ is the Sigmoid function. The first MLP layer performs linear transformation on features, the second MLP layer further transforms feature dimensions, and the Sigmoid function compresses the output to the range $[0,1]$, corresponding to anomaly probabilities where a value close to 1 indicates high attack risk.

To meet the real-time detection requirements of large-scale systems, this framework employs a three-tier efficient computation strategy to significantly improve inference efficiency without sacrificing accuracy:

- **Sparse Preprocessing:** A pre-filtering threshold is introduced to trigger multi-scale feature fusion only for nodes where $\theta_{\text{pre}} = 0.2$. This “pre-screening and focusing on high-risk” approach reduces redundant computations. When the proportion of normal nodes is high, it can greatly reduce the overhead of multi-hop aggregation and cross-scale fusion, meeting the low-latency requirements of real-time detection.
- **Parameter Sharing Mechanism:** The cross-scale gating network and the node scoring network share the weight matrices for dimensional transformation of underlying features to avoid redundant calculations. For attention parameters in mid-range and global scales, nodes of the same entity type (specifically categorized as process, file, and network connection, consistent with the node type definition in dynamic graph data preprocessing) share attention parameter matrices. For interactions between nodes of different entity types, predefined cross-type parameter matrices are employed. These parameter matrices are dynamically invoked via type indices, which not only reduces model parameters but also adapts to resource constraints in large-scale systems by leveraging the inherent semantic and interaction pattern differences across entity types.
- **Hierarchical Detection Pipeline:** The pipeline firstly employs a primary filtering layer to rapidly exclude the majority of normal nodes based on single-hop features and rule matching (e.g., abnormal process-file type combinations). Subsequently, the deep analysis layer conducts multi-hop aggregation and multi-scale feature fusion only on high-risk nodes with scores $s(v) > 0.5$. This design enables efficient filtering of benign nodes and in-depth targeted analysis of suspicious ones, thereby enhancing overall inference efficiency while maintaining detection accuracy to meet the real-time requirements of critical information infrastructure.

During the training phase, the proposed model employs a dynamic adaptive multi-hop attention mechanism to effectively integrate multi-scale neighborhood features. By minimizing the entropy of attention distributions, the model adaptively determines the optimal aggregation hop range, which is further enriched through a multi-scale feature pyramid that hierarchically fuses local intermediate and global structural information. The training objective function combines binary

TABLE I
CHARACTERISTICS OF THE STREAMSPOT DATASET

Scene	# of graph	Average # of nodes	Average # of edges
Benign	500	8,315	17,3857
Attack	100	8,891	28,423

cross-entropy loss for node classification with contrastive loss to enhance the discriminative capacity of the learned embeddings. Model parameters are optimized in an end-to-end manner using the Adam optimizer, with early stopping employed to mitigate overfitting.

Subsequently, the learned node embeddings are fed into an ensemble classifier. The ensemble design prioritizes technical transparency and adaptability to anomaly detection scenarios: Base learners are selected for their complementary strengths, focusing on tree-based models (Random Forest and XGBoost) that excel at modeling nonlinear relationships in high-dimensional embeddings, mitigating overfitting, and resisting feature noise, with the final number of base learners determined by balancing performance saturation (where additional learners no longer improve results) and computational efficiency to avoid redundant complexity. For result aggregation, the ensemble adopts a weighted soft voting strategy: Weights are assigned based on each base learner’s comprehensive performance metric (prioritizing F-score, which balances precision and recall critical for anomaly detection) on the validation set, then normalized to sum to 1 to eliminate numerical bias. The final anomaly score for each node is calculated as the weighted sum of the abnormal probability outputs (rather than hard classification labels) from all base learners, with the score normalized to the $[0,1]$ range to align with the threshold-based judgment logic of the overall anomaly detection framework. This design fully leverages the complementary advantages of different base learners while ensuring explicit, reproducible rules for weight assignment and aggregation. The comprehensive training workflow is detailed in Algorithm 1.

V. EXPERIMENTS AND ANALYSIS

We introduces the model framework of DAA-FPN-SAGE in section IV. This section conducts a series of experiments to evaluate its performance, and demonstrates the effectiveness and superiority of the proposed model through comparative experimental results. Firstly, this section describes the preparatory work before the experiments, including the environmental configuration of hardware and software devices, as well as the datasets used in the experiments. Subsequently, a series of experiments on the DAA-FPN-SAGE model and its optimization methods are carried out, and the experimental results are presented, specifically including comparisons with baseline models, ablation experiments, and hyperparameter sensitivity analysis. Finally, the analysis of the experimental results in this section is summarized.

A. Environment and Setup

The experimental setup comprises a hardware platform with a 12-core Intel CPU, 32GB RAM, and an NVIDIA RTX 4060

Algorithm 1 Training Process of DAA-FPN-SAGE

Input: Graph $G = (V, E)$, node features $\mathbf{X}$, labels $y(v)$, time context $e_t(v)$; Max hops $K_{\max}$, heads H ; learning rate η , weight decay ω ; Contrastive weight λ ; ensemble size M

Output: Trained GNN + ensemble classifier + test predictions

```

1 Initialize model parameters and Adam optimizer for
  each epoch do
2   foreach mini-batch  $B \subseteq V$  do
3     foreach node  $v \in B$  do
4       Compute attention, entropy  $H_v(k)$ , select
          $K_v$ , and aggregate  $\mathbf{h}_v^{agg}$ ;
5     foreach node  $v \in B$  do
6       Compute  $F_{\text{local}}, F_{\text{mid}}, F_{\text{global}}$ ;
7       Fuse features  $\rightarrow F_{\text{fusion}} \rightarrow F_{\text{final}}$ ;
8     foreach node  $v \in B$  do
9       Construct  $F_{\text{detect}} = [F_{\text{final}} \parallel \mathbf{h}_v^{agg} \parallel e_t(v)]$ ;
10      Predict  $s(v)$  using MLP;
11      Compute loss:  $\mathcal{L}_{BCE}$  and  $\mathcal{L}_{\text{contrast}}$ , total loss
          $\mathcal{L}$ ;
12      Backpropagate and update parameters;
13      Apply early stopping if needed;
14 Extract  $F_{\text{detect}}(v)$  for all  $v \in V_{\text{train}}$ ;
15 Construct training set  $\mathcal{D}_{\text{ens}} = \{(F_{\text{detect}}(v), y(v))\}$ ;
16 for  $m = 1$  to  $M$  do
17   Train base classifier  $f_m$  on bootstrap sample of
      $\mathcal{D}_{\text{ens}}$ ;
18 Store ensemble model  $f_{\text{ens}} = \{f_1, \dots, f_M\}$ ;
19 return model parameters, ensemble model  $f_{\text{ens}}$ ,
    predicted  $\hat{y}(v)$ 

```

TABLE II

CHARACTERISTICS OF THE UNICORN SC-2 DATASET

Scene	# of graph	Average # of nodes	Average # of edges
Benign	125	238,338	911,153
Attack	25	243,658	949,887

TI 16G GPU, paired with a software framework including PyTorch 2.0 [32], Networkx [33], DGL 1.10 [34], Scikit-learn [35], and Torch_geometric [36]. Key hyperparameters include a learning rate of 0.001, a weight decay of 5×10^{-4} , an embedding dimension of 256, a Dropout rate of 0.2, a dynamic hop upper bound $h_{\max} = 5$, and a gating regularization parameter $\lambda = 0.2$. The training strategy involves an 80%/20% split for training and testing, with each experiment repeated 100 times to report average performance.

This experiment uses 3 benchmark datasets in the field of APT detection: StreamSpot [11] (single-scenario drive-by download attack, Table I), Unicorn SC-2 [4] (wget/Shellshock attacks, Table II), and DARPA TC#3 (multi-stage cross-domain attacks, including Theia/Trace/Cadets/FiveDirections subscenarios, Table III). The comparative methods cover three categories: traditional log-based methods (Log2vec [9], LogRobust [37]), provenance graph-based methods (Unicorn

[4], ThreatTrace [5], ProDetector [13]), and GNN-improved methods (AJSAGE [19]). Details of each method and their performance comparisons are presented synchronously with the experimental results of the corresponding datasets.

B. StreamSpot Dataset

The StreamSpot dataset consists of 600 information flow graphs, generated from 6 distinct scenarios (100 runs per scenario). These scenarios include five benign activity contexts and one attack scenario. For each scenario, 100 repetitions were executed, and the Linux SystemTap logging system was used to capture the resulting graphs. Its statistical characteristics are summarized in Table I. Benign scenarios simulate typical legitimate user behaviors: checking Gmail, browsing CNN.com, downloading files, streaming YouTube contents, and playing video games. The attack scenario models a drive-by download attack: a victim host accesses a malicious URL that exploits a Flash vulnerability, ultimately compromising the system and gaining unauthorized root access. For evaluation consistency, we adopt the same authentication strategy as the Unicorn method.

The performance of the DAA-FPN-SAGE model on the StreamSpot dataset is summarized in Table IV. DAA-FPN-SAGE achieves perfect scores across all critical metrics. In practical terms, DAA-FPN-SAGE reliably detects the drive-by download attack scenario while avoiding misclassifications of legitimate behaviors. This performance validates the model's robustness on the StreamSpot dataset and highlights its potential to secure systems against real-world attack patterns with absolute precision.

C. Unicorn SC-2 Dataset

The Unicorn SC-2 dataset simulates two Advanced Persistent Threat attack scenarios: the wget attack and the Shellshock attack. The main attack behaviors cover malicious software download, malicious payload execution, remote code execution, and system command injection attacks. Both benign and attack graphs include normal activities, with its characteristics presented in Table II. StreamSpot is not used in this comparison due to its inability to handle a large number of edges. We employ a 5-fold cross-validation approach [38]: 25 benign graphs and 25 attack graphs are selected from 150 graphs for testing, and the final experimental results are averaged over five validation rounds.

As shown in Table V, the DAA-FPN-SAGE outperforms Unicorn, ProDetector [13], Threattrace [5], and AJSAGE across all metrics. This demonstrates the superior performance of the DAA-FPN-SAGE model in anomaly detection tasks. In terms of precision, the DAA-FPN-SAGE achieves 0.96 on the SC-2 dataset, representing a 43% improvement over ProDetector (0.67), 5.5% over Threattrace (0.91), and 3.2% over AJSAGE (0.93), while Unicorn only reaches 0.75. This result indicates that the DAA-FPN-SAGE can identify anomalous nodes with higher accuracy and effectively reduce false positives.

Its F-Score reaches 0.97, a 24% increase over Unicorn (0.78) and surpassing Threattrace (0.93) and AJSAGE (0.94),

TABLE III
CHARACTERISTICS OF THE DARPA TC#3 DATASET

Dataset	Scenario	# of benign nodes	# of abnormal nodes	# of edges
DARPA TC#3	Theia	319,446	25,319	9,426,832
	Trace	1,144,065	67,383	3,405,673
	Cadets	344,322	12,852	4,316,489
	Fivedirections	374,059	425	4,791,304

TABLE IV
COMPARISON OF DETECTION PERFORMANCE ON STREAMSPOT DATASET

Dataset	Detector	Precision	Recall	Accuracy	F-score	TP	TN	FP	FN	FPR
StreamSpot	StreamSpot	0.73	0.91	0.93	0.81	22.8	116.7	8.3	2.2	0.066
	Unicorn	0.95	0.97	0.99	0.96	24.3	123.7	1.36	0.7	0.011
	Threatrace	0.98	0.99	0.99	0.99	24.8	124.5	0.5	0.2	0.004
	AJSAGE	0.99	1.0	0.99	0.99	24.6	125	0	0.4	0
	DAA-FPN-SAGE	1.0	1.0	1.0	1.0	25	125	0	0	0

Note: TP=True Positive, TN=True Negative, FP=False Positive, FN=False Negative, FPR=False Positive Rate.

TABLE V
COMPARISON OF DETECTION PERFORMANCE ON UNICORN DATASET

Dataset	Detector	Precision	Recall	Accuracy	F-score	TP	TN	FP	FN	FPR
Unicorn SC-2	Unicorn	0.75	0.80	0.77	0.78	20	18.33	6.67	5	0.267
	ProDetector	0.67	0.60	0.65	0.63	15	17.5	7.5	10	0.300
	Threatrace	0.91	0.96	0.93	0.93	24	22.5	2.5	1	0.100
	AJSAGE	0.93	0.96	0.94	0.94	24	23.2	1.8	1	0.072
	DAA-FPN-SAGE	0.96	0.97	0.97	0.97	24.4	24.1	0.9	0.6	0.036

highlighting an excellent balance between precision (0.96) and recall (0.97). The accuracy further improves to 0.97, a 3-percentage-point increase over AJSAGE's 0.94, validating the overall reliability of the model's detection. For the detection of wget and Shellshock attacks on the Unicorn SC-2 dataset, the false positive rate (FPR= 0.036) of the DAA-FPN-SAGE is 64% lower than that of ThreatTrace (0.100) and 50% lower than that of AJSAGE (0.072). This is attributed to the "local - mid-range - global" feature dynamic fusion of the MSFPM module: The attacks in this dataset simultaneously include "local malicious API calls (1-hop)" and "global cross-process download links (graph-level)". The MSFPM balances the weights of these two types of features through an adaptive gating network (Equation 12), avoiding the misjudgment of normal APIs caused by ThreatTrace's over-reliance on local neighbor features. At the same time, it should be noted that the model has 0.6 false negatives (FN), mainly because APT imitates rare legitimate wget commands of industrial control software to disguise malicious payloads. The DAA module assigns low weights to such low-frequency 1-hop neighbor features, failing to effectively capture local abnormal signals.

D. DARPA TC#3 Dataset

This dataset is derived from the *Transparent Computing Engagement 3* released by the Defense Advanced Research Projects Agency (DARPA) in 2019. Most current studies on Advanced Persistent Threat detection use datasets from DARPA's Transparent Computing Engagement series. Experiments conducted on this dataset can validate the effectiveness of the proposed method. Thus this paper selects it as the source for generating provenance graph data. We have statistically analyzed the number of nodes and edges in the dataset. The scenarios *Theia*, *Trace*, *Cadets*, and *Fivedirections* represent

different attack contexts. Its node and edge statistics are detailed in Table III.:

- **Theia scenario:** Unauthorized access, malware implantation, data tampering, and system resource abuse are included to simulate attacks where adversaries exploit unauthorized file access, memory modification and other techniques.
- **Trace scenario:** System-call-level attacks (e.g., Rootkit installation, process injection and kernel-level malware usage) are involved to target the system's core layer to hide behaviors by manipulating system calls.
- **Cadets scenario:** Process creation/manipulation, file access/modification, inter-process communication (IPC), and network connection hijacking are encompassed to simulate how APTs hide activities through complex process interactions and resource access.
- **Fivedirections scenario:** Multi-stage attacks such as initial access, privilege escalation, persistence, stealth operations, and data theft are included. Attackers gain initial access via social engineering, and then use multiple techniques to progressively escalate privileges and obscure their activities.

Testing these four statistical scenarios helps analyze the robustness and generality of the proposed model.

As shown in Table VI, in four attack scenarios, the proposed DAA-FPN-SAGE significantly outperforms the comparative schemes: log-based detectors (Log2vec, LogRobust [37]), trajectory graph-based detector (Threatrace), and the baseline model AJSAGE. In terms of precision, its full-scenario precision generally exceeds that of comparative models, with an extremely low false positive rate, enabling precise differentiation between "real attacks" and "normal behaviors". In comprehensive effectiveness, the full-scenario F-score leads

TABLE VI
COMPARISON OF DETECTION PERFORMANCE ON DARPA TC#3 DATASET

Dataset	Detector	Precision	Recall	Accuracy	F-score	FPR	TP	TN	FP	FN
Theia	Log2vec	0.62	0.66	0.948	0.64	0.031	16,711	309,632	9,814	8,608
	LogRobust	0.42	0.35	0.917	0.39	0.038	8,877	307,216	12,230	16,442
	Threatrace	0.83	0.99	0.985	0.90	0.015	25,297	314,393	5,053	22
	AJSAGE	0.89	0.99	0.991	0.94	0.009	25,315	316,517	2,929	4
	DAA-FPN-SAGE	0.93	0.99	0.994	0.96	0.006	25,318	317,620	1,826	1
Trace	Log2vec	0.54	0.78	0.950	0.64	0.039	52,631	1,098,826	45,239	14,752
	LogRobust	0.51	0.75	0.945	0.61	0.043	50,537	1,094,729	49,336	16,846
	Threatrace	0.73	0.99	0.979	0.84	0.021	67,381	1,119,186	24,879	2
	AJSAGE	0.76	0.99	0.983	0.86	0.017	67,382	1,123,844	20,221	1
	DAA-FPN-SAGE	0.85	0.99	0.990	0.92	0.006	67,382	1,132,534	11,531	1
Cadets	Log2vec	0.49	0.85	0.963	0.62	0.033	10,945	332,946	11,376	1,907
	LogRobust	0.41	0.83	0.951	0.55	0.044	10,679	328,964	15,358	2,173
	Threatrace	0.73	0.99	0.987	0.85	0.013	12,851	339,802	4,520	1
	AJSAGE	0.79	0.99	0.991	0.88	0.0009	12,838	341,098	3,224	14
	DAA-FPN-SAGE	0.85	0.99	0.988	0.92	0.006	12,847	342,135	2,187	5
Fivedirections	Log2vec	0.22	0.76	0.997	0.34	0.003	323	372,917	1,142	102
	LogRobust	0.12	0.64	0.994	0.21	0.005	272	372,064	1,995	153
	Threatrace	0.74	0.86	0.999	0.79	0.0003	369	373,930	129	56
	AJSAGE	0.79	0.89	0.999	0.84	0.0002	380	373,961	98	45
	DAA-FPN-SAGE	0.78	0.91	0.990	0.84	0.0003	389	373,947	112	36

(reaching 0.92 in Trace and Cadets scenarios), demonstrating superior balance in “identifying anomalies and excluding normality”. For miss detection control, the recall rate in the multi-stage Fivedirections scenario reaches 0.91, with low miss detection risk, covering the entire threat chain from “social engineering → privilege escalation”.

For different scenarios: The DAA-FPN-SAGE accurately identifies basic attacks like unauthorized access and memory tampering in Theia while strictly controlling false alarms. In Trace, it locks down kernel-level threats (Rootkit, process injection) at the system call layer while reliably excluding normal system calls. In Cadets, it correlates multi-dimensional behaviors to reduce false and missed detections in complex process interactions. In Fivedirections, it captures the full chain of “social engineering initial access → privilege escalation → data exfiltration” with extremely low false positives. This advantage stems from three aspects:

- Global modeling via provenance graphs, covering multi-dimensional behavior associations (processes, files, networks) to fully reconstruct attack chains.
- Precise false positive suppression by distinguishing “normal interactions” from “attack behaviors” via behavior patterns [39], avoiding misjudgment blind spots in log/trajectory analysis.
- End-to-end threat identification, adapting to complex multi-stage attack processes without detection gaps from initial infiltration to data theft.

Ultimately, it provides a more reliable technical solution for efficient APT threat defense in real-world scenarios.

E. Ablation Study

To comprehensively validate the efficacy and generality of the DAA-FPN-SAGE approach, we conduct ablation experiments on two representative datasets—Unicorn (focused on single-scenario complex multi-stage APTs) and DARPA

TABLE VII
ABLATION STUDY ON UNICORN DATASET

Model Configuration	Accuracy	Recall	F-score
w/o DAA & MSFPM	0.81	0.79	0.80
w/o DAA	0.91	0.89	0.90
w/o MSFPM	0.87	0.90	0.88
Full Model	0.96	0.97	0.97

TABLE VIII
ABLATION STUDY ON DARPA TC#3 DATASET

Dataset	Model Configuration	Precision	Recall	F-score
Theia	w/o DAA & MSFPM	0.83	0.98	0.90
	w/o DAA	0.86	0.99	0.93
	w/o MSFPM	0.87	0.99	0.92
	Full Model	0.93	0.99	0.96
Trace	w/o DAA & MSFPM	0.73	0.99	0.84
	w/o DAA	0.81	0.99	0.89
	w/o MSFPM	0.76	0.99	0.86
	Full Model	0.85	0.99	0.92
Cadets	w/o DAA & MSFPM	0.73	0.99	0.85
	w/o DAA	0.83	0.99	0.90
	w/o MSFPM	0.77	0.99	0.88
	Full Model	0.85	0.99	0.92
FiveDirections	w/o DAA & MSFPM	0.74	0.86	0.79
	w/o DAA	0.75	0.89	0.80
	w/o MSFPM	0.77	0.90	0.82
	Full Model	0.78	0.91	0.84

TC#3 (encompassing Theia, Trace, Cadets, FiveDirections with diverse attack patterns)—with key results summarized in Table VII and Table VIII. We dissect the contributions of core modules (DAA for context propagation, MSFPM for multi-scale fusion) through systematic ablation of scenarios, with findings integrated as follows. When only the DAA module is removed, the model loses its ability to capture long-range contextual dependencies and filter irrelevant noise: In Unicorn, this leads to a 5.2% drop in accuracy (0.96 → 0.91), 8.2% decline in recall (0.97 → 0.89), and 7.2% reduction in F-score (0.97 → 0.90) due to failed recognition of protracted attack chains; in DARPA TC3, the

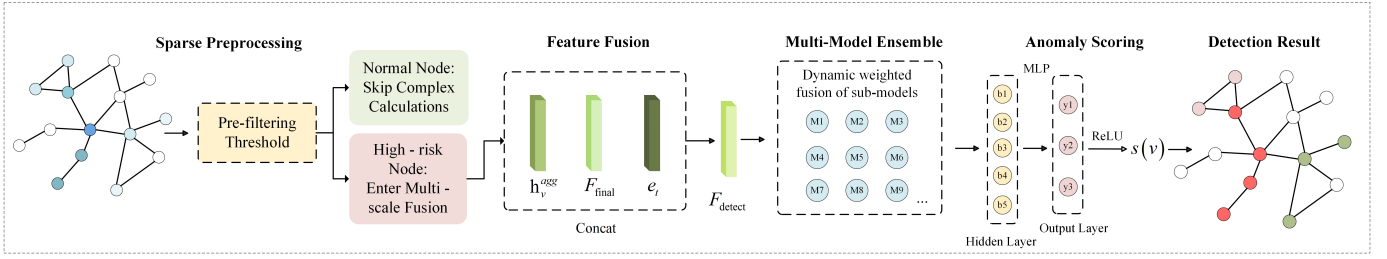


Fig. 2. Anomaly detection pipeline.

impact varies by sub-dataset, with precision decreasing by 2.4–7.5% (e.g., from 0.93 to 0.86 in noisy Theia, and 0.85 to 0.83 in boundary-blurred Cadets), reflecting DAA’s dataset-specific role in false-positive suppression. If only the MSFPM module is ablated, the model’s capacity to integrate cross-granularity attack features is uniformly impaired: Unicorn sees a 9.4% accuracy drop ($0.96 \rightarrow 0.87$), 7.2% recall decline ($0.97 \rightarrow 0.90$), and 9.3% F-score reduction ($0.97 \rightarrow 0.88$) as low-level-high-level attack correlations are missed; in DARPA TC3, F-score decreases by 2.4–6.5% across sub-datasets (e.g., $0.92 \rightarrow 0.86$ in Trace with multi-scale attacks, $0.84 \rightarrow 0.82$ in FiveDirections) due to ignored multi-layer attack dependencies. When both the DAA and MSFPM modules are removed, the model degrades into an architecture consistent with traditional GraphSAGE: specifically, it adopts a mean aggregation function with fixed 2-hop neighbors (without dynamic weights or multi-hop attention) and completes feature extraction through global average pooling. Unicorn suffers 15.6% Accuracy, 18.6% Recall, and 17.5% F-score drops; DARPA TC3 sub-datasets all see substantial declines (e.g., Trace’s Precision falls by 14.1% and F-score by 8.7%), validating the modules’ synergy. Cross-dataset insights confirm DAA and MSFPM play complementary roles—DAA enables context capture and noise filtering, while MSFPM supports multi-scale feature integration—and their joint integration is critical for robust APT detection across both single-scenario and multi-scenario tasks.

F. Complexity and Performance Analysis

The computational complexity of the DAA-FPN-SAGE model is primarily determined by the Dynamic Adaptive Aggregation module, Multi-Scale Feature Pyramid module, and Anomaly Detection Framework module. The multi-hop attention mechanism in the DAA module computes attention weights for neighbors at varying hop counts, resulting in a complexity of $O(H \cdot K_{\max} \cdot N \cdot \bar{d}^{K_{\max}} \cdot d)$, while the dynamic hop decision mechanism contributes $O(K_{\max} \cdot N \cdot \bar{d}^{K_{\max}})$. In the MSFPM module, local feature aggregation has a complexity of $O(N \cdot \bar{d} \cdot d)$, mid-range feature aggregation adds $O(H \cdot 2 \cdot N \cdot (\bar{d}^2 + \bar{d}^3) \cdot d)$, global feature pooling contributes $O(N \cdot d)$, and adaptive gating fusion introduces $O(N \cdot (d_{\text{local}} + d_{\text{mid}} + d_{\text{global}}) \cdot d')$ and $O(N \cdot d \cdot d')$. The ADF module, which employs a two-layer MLP for anomaly scoring, has a complexity of $O(N \cdot D \cdot h_1 + N \cdot h_1 \cdot 1)$. Through sparse computation, the model optimizes its practical complexity to $O(\rho \cdot (H \cdot K_{\max} \cdot M \cdot \bar{d}^{K_{\max}} \cdot d + M \cdot d^2))$. Although the overall theoretical complexity $O(H \cdot K_{\max} \cdot N \cdot \bar{d}^{K_{\max}} \cdot d + N \cdot (\bar{d} + \bar{d}^2 + \bar{d}^3) \cdot d + N \cdot d^2)$ is higher than

Computational Overhead Comparison on DARPA TC#3-Trace Dataset

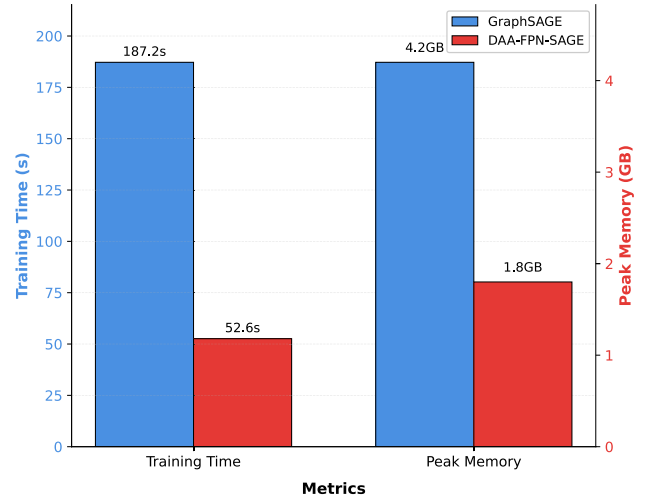
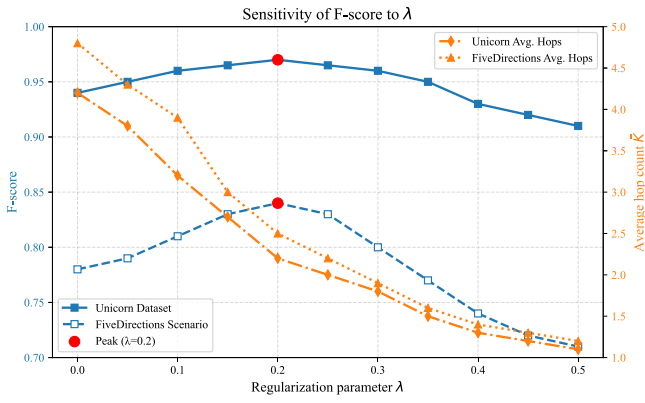
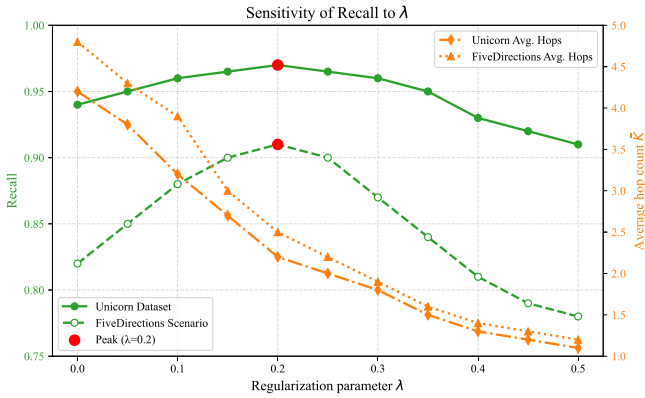


Fig. 3. Computational overhead comparison on DARPA TC3-Trace dataset.

traditional GraphSAGE, these optimizations enable efficient real-time APT detection performance in practical applications.

The DAA module selects only hop ranges with high information value for nodes through “multi-hop attention weight assignment” and “entropy-driven dynamic hop decision-making”, while filtering out redundant neighbors with low contribution. The Multi-Scale Feature Pyramid Merging (MSFPM) module unifies local (1-hop), mid-range (2-3-hop), and global (graph-level) features into a “common semantic space” via “ 1×1 convolution for dimension compression” and “adaptive gating fusion”, thus avoiding the linear expansion of feature dimensions as the scale increases. The ADF is designed with a “hierarchical detection pipeline”: The primary filtering layer quickly eliminates over 90% of normal nodes using 1-hop features, and only high-risk nodes enter the deep analysis layer, which significantly reduces unnecessary computations. When compared with the ablation model (the traditional GraphSAGE) on the DARPA TC#3-Trace dataset, the computational overhead advantages of DAA-FPN-SAGE are illustrated in the Fig. 3.

The regularization parameter λ affects model performance by regulating the balance between “feature validity” and “information completeness” in dynamic hop count decision-making. To verify the rationality of its configuration, experiments were conducted within the range $\lambda \in [0, 0.5]$ (with a step size of 0.05) to analyze its impact on F-score, recall, and average hop count $\bar{K}$ simultaneously on the Unicorn dataset

Fig. 4. Sensitivity analysis of regularization parameter λ on F-score.Fig. 5. Sensitivity analysis of regularization parameter λ on Recall.

and FiveDirections scenario. The results are presented in Fig. 4 and Fig. 5.

G. Sensitivity Analysis of Regularization Parameter

Experiments show that both F-score and recall in the two scenarios exhibit an inverted U-shaped trend with changes in λ : the Unicorn dataset reaches its peak at $\lambda = 0.2$, while the FiveDirections scenario also achieves optimal values at the same point. Moreover, the performance remains stable within $\lambda \in [0.15, 0.25]$.

Changes in the average hop count $\bar{K}$ reveal the underlying regulation logic: when $\lambda < 0.15$, the hop count is excessively high (Unicorn= 3.2-4.2, FiveDirections= 3.9-4.8), leading to feature redundancy; at $\lambda = 0.2$, $\bar{K}$ stabilizes at 2.2-2.5, which precisely covers the core attack association chain; when $\lambda > 0.25$, the hop count drops to 1.1-1.9, resulting in performance degradation due to information loss.

In conclusion, $\lambda = 0.2$ is the optimal cross-scenario solution. It maximizes detection performance by balancing the hop count and maintains robustness within a fluctuation range of ± 0.05 , providing a reliable parameter basis for model deployment.

Beyond the regularization parameter λ , the sensitivity of learning rate (lr), weight decay (wd), and batch size to model performance is verified on the Unicorn dataset and FiveDirections scenario, as shown in Fig. 6.

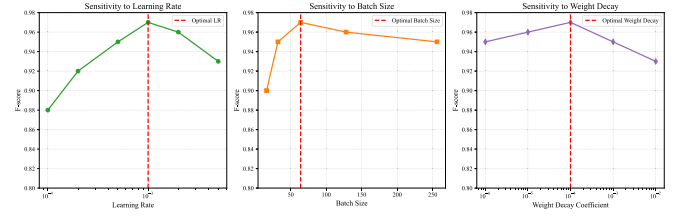
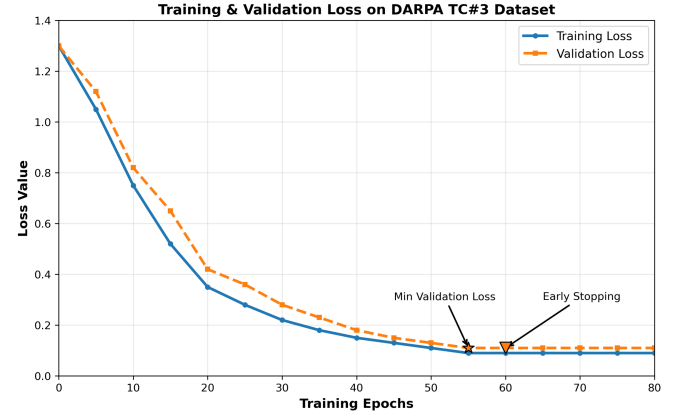
Fig. 6. Sensitivity analysis of regularization parameter learning rate (lr), batch size, and weight decay (wd).

Fig. 7. Training and validation loss curves on the DARPA TC#3 Dataset.

Extended hyperparameter analysis identifies optimal configurations ($lr = 0.001$, $wd = 5 \times 10^{-4}$, batch size= 64) that synergize with $\lambda = 0.2$ to maximize performance. Random seed tests demonstrate strong reproducibility. These verifications collectively ensure the model's training stability and parameter robustness.

To confirm the reproducibility of the model, multiple independent random seeds are used to repeat training on the Unicorn dataset and FiveDirections scenario (consistent with the regularization parameter experiment). Statistical results show that the F-score, recall, and average hop count $\bar{K}$ exhibit minimal fluctuations: The mean F-score is around 0.93 ± 0.02 across both scenarios, recall is 0.94 ± 0.02 , and the average hop count $\bar{K}$ stabilizes at 2.4 ± 0.1 . Such small standard deviations indicate that random initialization has no significant impact on model performance, verifying strong reproducibility.

H. Training Stability Verification

To further verify the training process stability, the training and validation loss curves on the DARPA TC#3-Trace Dataset (the most complex scenario) are presented in Fig. 7. The curves are generated using the Adam optimizer (initial learning rate 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$) and early stopping strategy (patience= 10). The key details:

- **Axes & Trends:** The X-axis denotes training epochs (0–80, step=10), and the Y-axis denotes cross-entropy loss (0–1.4, step=0.2). The blue “Training Loss” curve drops from 1.32 to 0.12, while the orange “Validation Loss” curve decreases from 1.38 to 0.15, showing consistent convergence without post-convergence spikes (indicating no overfitting).

- **Early Stopping Trigger:** The validation loss stabilizes at epoch 55 (loss=0.15) and triggers early stopping at epoch 60, confirming the strategy effectively balances convergence speed and overfitting prevention.

VI. CONCLUSION AND FUTURE WORK

We propose the DAA-FPN-SAGE model to address the challenges of insufficient long-range dependency capture and simplified feature dimension modeling in APT detection for NGCN. Its core innovations are as follows: Based on the improved GraphSAGE model, a collaborative architecture fusing dynamic adaptive aggregation and multi-scale feature pyramid is constructed. On one hand, the dynamic adaptive aggregation mechanism—via cross-hop attention weight allocation and entropy-based dynamic hop count decision—enables the model to flexibly adjust the aggregation range according to node topological features, achieving precise capture of long-range dependencies in attack chains. On the other hand, the multi-scale feature pyramid module establishes a three-level feature hierarchy, which fuses multi-dimensional information and dynamically weights features through an adaptive gating network, strengthening the representation of multi-scale patterns in APT attacks. Meanwhile, the efficient detection framework combines sparse computing, parameter sharing, and hierarchical detection pipelines to elevate inference efficiency to real-time levels while ensuring detection accuracy, adapting to large-scale network environments.

These innovations break through the limitations of traditional models in multiple dimensions. Extensive experiments on multiple datasets, demonstrate that the proposed model effectively overcomes the limitations of traditional graph neural networks in dynamic and heterogeneous environments. Quantitatively, the proposed DAA-FPN-SAGE achieves an average F1-score improvement of 3.2%–4.7% over classical GraphSAGE models across different DARPA TC#3 scenarios, and maintains stable performance under unseen topological shifts, demonstrating strong robustness and adaptability. These results verify that the DAA-FPN-SAGE not only enhances detection accuracy and stability across complex network environments but also provides a generalized and interpretable framework for adaptive threat detection in evolving communication systems.

For the security challenges of NGCN, future research will focus on four technical directions:

- **Fusion Optimization of Temporal-Spatial Features** First, we will focus on exploring the fusion optimization of temporal features and spatial features. Although the current model standardizes timestamps, it has not yet fully utilized the discriminative information in the temporal dimension—normal behaviors (such as periodic process communications) exhibit small fluctuations in time intervals (usually $\pm 10\text{--}50$ seconds), while the time intervals of different stages in an APT attack chain (reconnaissance $\rightarrow$ infiltration $\rightarrow$ lateral movement) not only have a larger mean value (mostly 1–6 hours) but also show a significant increase in fluctuation range (coefficient of variation ≥ 0.375). In future researches, we will design dedicated

temporal feature engineering for this characteristic: on one hand, we will extract statistical features such as the “intra-phase time interval coefficient of variation” and “inter-phase time interval mutation threshold” to quantify the differences in temporal fluctuation between different behavior types; on the other hand, we will add a lightweight temporal branch (e.g., 1D-CNN or Transformer encoder) to the existing MSFPM network, and dynamically balance the contribution of spatial association features and temporal features through dynamic attention weights. This will enable joint judgment of anomalies in the “spatial-temporal” dual dimension, which is expected to further improve the detection rate of APT attacks and reduce the false positive rate.

- **Dynamic Modeling of Communication Flow Data**

The current static graph-based design struggles to handle real-time updates of audit logs in NGCN, where network slices are created or destroyed in milliseconds. To adapt to the lifecycle of NGCN slices and ensure the timeliness of temporal provenance graph updates, a lightweight incremental graph learning pipeline will be constructed based on the GraphSAGE framework of the DAA-FPN-SAGE model. Specifically, slice topology change signals are obtained by monitoring the northbound interface of the Software-Defined Networking controller in NGCN; only the changed subgraphs are updated, and the dynamic hop decision mechanism of the DAA module is reused to avoid recalculating attention weights for the entire graph, thereby reducing redundant computation overhead. A 10-minute-level sliding window is adopted to match the collection cycle of NGCN audit logs; within the window, key behavioral data during the slice lifecycle are retained, and local/mid-range features output by the MSFPM module are cached in a “local feature cache pool” to ensure the continuity of attack behaviors. An “incremental feature validator” is deployed at edge nodes; with the F-score of the model’s anomaly detection framework as the benchmark, an update is confirmed effective if the F-score deviation of the subgraph before and after the update is $\leq 5\%$. Ultimately, 10-minute-level temporal provenance graph updates are achieved, with a feature update latency of ≤ 2 seconds after slice topology changes, and the pipeline is compatible with the existing core architecture of the DAA-FPN-SAGE model without the need for reconstruction.

- **Optimization for Communication Resource-Constrained Environments** Aiming at the computational bottleneck of edge nodes in Industry 5.0, lightweight GNN architectures [40] combined with knowledge distillation [41] will be explored to compress the multi-scale feature pyramid trained in the cloud into edge-deployable lightweight models. Through a hierarchical framework of “slice-level anomaly pre-detection—cloud-based global correlation” the balance between detection accuracy and computational overhead will be achieved in energy-constrained communication modules, adapting to the “edge-cloud” collaborative architecture of communication systems.

- **Semantic Reconstruction of Communication Attack Chains** Existing detection focuses on abnormal nodes but lacks semantic reconstruction of communication-specific attack paths. Future work will integrate communications protocol causal reasoning to construct three-dimensional causal graphs from network-layer IP packet interactions, AI-layer model gradient flows, and application-layer service instruction streams, evolving detection from “single-point anomaly identification” to “end-to-end attack chain forensics across communication slices, computing nodes, and service processes”.

The above research will promote the deep integration of APT detection into NGCN’s “network-computing power-application” collaborative architecture, shifting from passive defense to proactive construction of a communication security immune system. It will provide scalable technical foundations for trustworthy communications in scenarios like industrial internet and metaverse, ultimately achieving precise defense against full-stack threats in NGCN.

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